

Portable, Extensible SCADA

A modern supervisory control platform for any process built from sensors, actuators, controllers, electrical panels, mechanical equipment, and field devices. OptiPanel centralizes monitoring, control, alarms, history, reports, automation, and web HMI engineering in one deployable system.

- WEB HMI
- HISTORIAN
- ALARMS
- AUTOMATION
- OPEN INTEGRATION

SCOPE

Production plants, utilities, building management, OEM machines, facilities, and infrastructure

CORE

Live acquisition, supervision, control, event handling, and operating history

DELIVERY

Browser runtime, server deployment, Docker, headless mode, and desktop packages

EXTENSIBILITY

Protocol plugins, REST APIs, scripts, scheduler, reports, and Node-RED workflows

SUPERVISORY CONTROL LOOP

Sense

Read process values, status bits, counters, alarms, measurements, and states from field equipment.

Supervise

Normalize tags, evaluate limits, monitor devices, manage quality, and distribute current state.

Control

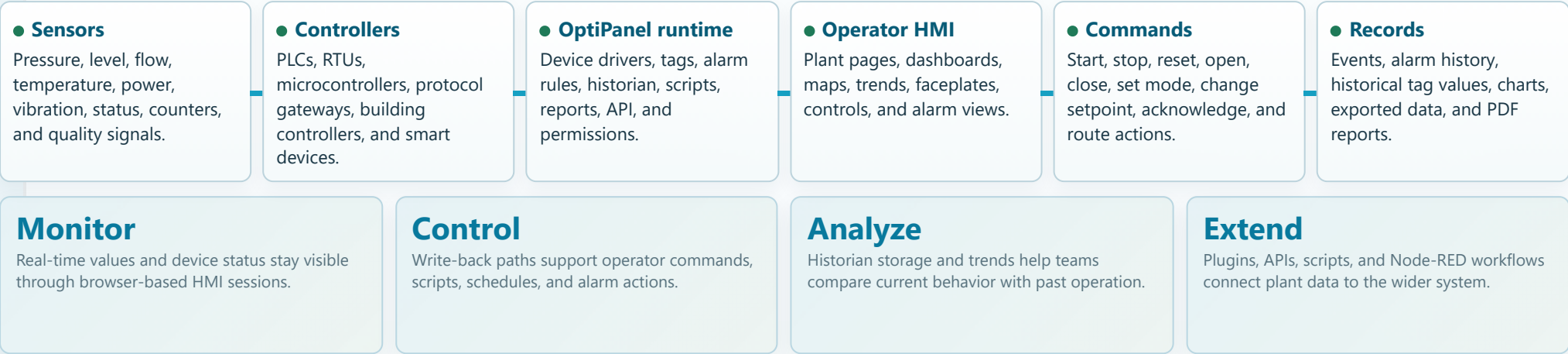
Write commands to valves, pumps, drives, dampers, breakers, setpoints, modes, and interlocks when authorized.

Record

Store trends, alarms, events, reports, audit context, and operating records for review and improvement.

Central monitoring and control for real-world processes.

SCADA sits above field devices and controllers. It gathers data, presents operating context, records process history, alerts people when conditions change, and sends controlled commands back to equipment. OptiPanel is built for that supervisory layer.



Any process with sensors and actuators can become an OptiPanel application: machines, treatment plants, HVAC systems, energy rooms, production lines, tanks, pumps, valves, drives, meters, and electro-mechanical assets.

SCADA ROLE	WHAT OPTIPANEL PROVIDES	OPERATIONAL RESULT
Data acquisition	Protocol devices, polling, status tracking, reconnect behavior, browsing, tag get/set, and socket updates.	Operators see current state from different equipment families in one application.
Supervisory control	HMI commands, script functions, scheduler writes, alarm actions, and permission-aware controls.	Control actions can be structured, visible, and aligned with operating authority.
Operating memory	DAQ historian, current values database, event history, alarm history, charts, and PDF reports.	Process behavior becomes reviewable instead of disappearing after the screen changes.

One SCADA platform, many industrial and infrastructure domains.

OptiPanel is not limited to one demo project. The same platform model applies wherever field data must be monitored centrally and equipment must be controlled safely.

PRODUCTION PLANT
Line status, machine cells, conveyors, utilities, batch support, downtime reasons, energy meters, shift dashboards, and line-level controls.

BUILDING MANAGEMENT
HVAC, chilled water, pumps, fans, dampers, room conditions, electrical panels, alarms, tenant areas, and scheduled operation.

WATER AND UTILITIES
Pumps, valves, tanks, pressure, flow, chemical dosing, reservoirs, lift stations, treatment stages, alarms, and compliance reporting.

ENERGY AND FACILITIES
Generators, switchgear status, meters, power quality, compressors, boilers, storage, environmental monitoring, and maintenance workflows.

OEM EQUIPMENT
Packaged machines with embedded HMI, recipe-like settings, device diagnostics, remote dashboards, and commissioning screens.

FOOD AND PROCESS
Tanks, mixers, CIP utilities, temperature trends, production zones, sanitation states, alarms, history, and report packs.

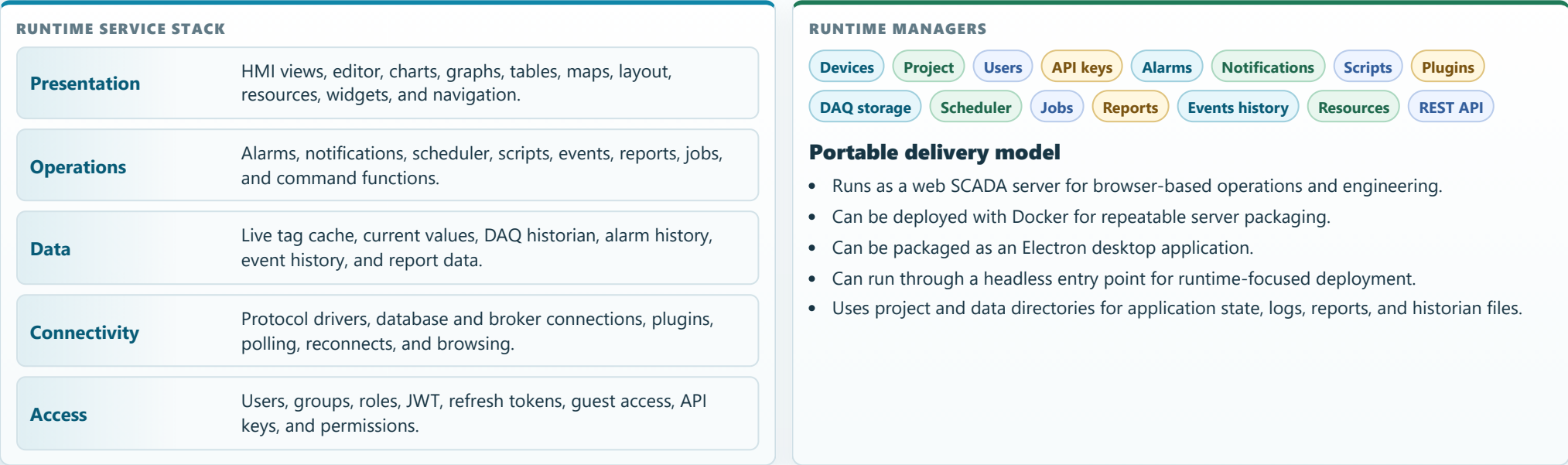
LOGISTICS AND MATERIAL HANDLING
Conveyors, sorters, barcode events, gates, counters, machine status, exception alarms, and map-based area views.

REMOTE ASSETS
Distributed stations, local gateways, MQTT/HTTP sources, maps, historical review, notifications, and browser access.

APPLICATION PATTERN	TYPICAL FIELD ELEMENTS	OPTIPANEL MODULES INVOLVED
Machine monitoring and control	Sensors, motors, drives, valves, cylinders, counters, status bits, emergency signals.	Drivers, tags, HMI views, controls, alarms, scripts, trends, events.
Plant-wide supervision	Multiple PLCs, process areas, utility systems, dashboards, role-specific navigation.	Project model, views, layout, user roles, historian, reports, maps.
Building automation	BACnet devices, HVAC units, fans, dampers, pumps, meters, schedules, zones.	BACnet, scheduler, maps, dashboards, alarms, notifications, users.
Data integration layer	Databases, Redis, MQTT brokers, HTTP APIs, Node-RED flows, external reporting systems.	ODBC, Redis, MQTT, Web API, Node-RED, REST API, API keys.

A portable web SCADA architecture built around devices, tags, services, and HMI sessions.

The platform combines a Node.js runtime, an Angular web interface, Socket.IO live communication, REST APIs, project storage, plugin-aware connectivity, and runtime managers for operations services.



LAYER	PLATFORM RESPONSIBILITY	WHY IT MATTERS
Field layer	PLC, RTU, sensor, actuator, database, broker, web service, and controller communication.	OptiPanel can sit above mixed plant equipment instead of requiring one vendor stack.
Tag layer	Typed values, units, scaling, expressions, deadband, DAQ settings, scripts, and write behavior.	HMI, alarms, reports, scripts, and integrations work from the same point model.
Service layer	Alarms, history, events, scheduler, notifications, reports, API, security, and plugins.	The system becomes a complete operations platform, not a static screen builder.

Industrial and IT connectivity through built-in drivers and installable plugins.

OptiPanel supports field protocols, database access, broker-style integrations, web requests, and platform-to-platform communication. Connection drivers normalize these sources into tags that the rest of the SCADA system can use.

CONNECTION FAMILIES

OPC UA

Siemens S7

Modbus TCP

Modbus RTU

BACnet

MQTT

Web API / HTTP

EtherNet/IP PCCC

Beckhoff ADS

Mitsubishi MELSEC

ODBC

Redis

GPIO

Webcam

OptiPanel Server

PLC AND CONTROLLER LINKS

Siemens S7, OPC UA, Modbus TCP/RTU, EtherNet/IP PCCC, Beckhoff ADS, Mitsubishi MELSEC, and GPIO support process and machine control projects.

BUILDING AND FACILITY LINKS

BACnet connects building controllers and HVAC points while Modbus and OPC UA cover many meters, drives, and utility devices.

SOFTWARE AND DATA LINKS

MQTT, HTTP/Web API, ODBC, Redis, OptiPanel Server, and Node-RED integration connect operational data with software systems.

CAPABILITY	DETAILS	RESULT
Device lifecycle	Start, stop, restart, enable, connection status, reconnect behavior, polling, browsing where supported.	Engineering and operations can understand whether the equipment link is healthy.
Data browsing	OPC UA, MQTT, ODBC, Redis, BACnet, and HTTP paths expose browser or discovery operations where available.	Faster tag setup and easier commissioning for supported data sources.
Write-back	Tag writes flow through device-specific setValue implementations and runtime command functions.	Operator buttons and automation can command real process points when configured.
Plugin model	Server plugin registry includes connection devices, connection databases, chart/report support, and services.	The platform can be extended without reducing the whole product to one fixed driver set.

The tag model links field values to screens, controls, alarms, reports, and automation.

A SCADA point is not only an address. OptiPanel tags carry engineering metadata and runtime behavior so one source of truth can serve HMI graphics, alarms, DAQ, commands, scripts, and integrations.

IDENTITY AND ADDRESS
Device, tag ID, friendly name, address, variable mapping, format, type, direction, and initial value.

ENGINEERING MEANING
Units, min/max, decimals, scaling, date/time conversion, and value formatting for operator readability.

SIGNAL BEHAVIOR
Deadband, edge handling, read/write expressions, scripts, restore behavior, and DAQ options.

OPERATIONAL USE
HMI bindings, alarm evaluation, historical storage, reports, scheduler writes, scripts, and API commands.

CONTROL EXAMPLES

- Valve open/close command tags with feedback status and interlock indication.
- Pump start/stop command tags with running, fault, auto/manual, and trip state.
- Setpoint write tags for pressure, temperature, flow, speed, level, or operating mode.
- Momentary reset, acknowledge, and sequence command points.

SUPERVISORY EXAMPLES

- Analog limits used by alarms, charts, reports, and dashboard summaries.
- Calculated values from read expressions or scripts.
- Runtime state retained in current storage for last-known-value recovery.
- Permission context that controls who can see or operate a point.

TAG FUNCTION	WHERE IT APPEARS	WHY IT MATTERS
Live value	HMI objects, dashboards, charts, tables, scripts, Node-RED nodes, and REST command endpoints.	One point can feed many operating experiences without duplicate logic.
Write command	HMI controls, scripts, scheduler, alarm actions, Node-RED, and API access.	Control is consistent, traceable, and easier to protect with authorization.
Historian point	DAQ storage, trend charts, report tables, and historical queries.	Trend and report behavior is linked to the engineering definition of the point.

Web HMI pages for operating real equipment, not static presentation drawings.

OptiPanel includes a browser-based engineering and operating environment for building process graphics, dashboards, maps, charts, tables, controls, and multi-page HMI systems.

VIEWS EDITOR
Build process views, equipment pages, panels, cards, grouped elements, and multi-view containers.

PROCESS SYMBOLS
Use SVG editing, custom styles, resources, widgets, pumps, valves, tanks, controls, and imported graphics.

LIVE OBJECTS
Bind text, gauges, charts, tables, indicators, sliders, buttons, and animated pipes to tag values.

NAVIGATION
Header bar, side menu, start view, cards, maps, language display, login overlay, zoom, and auto resize.

OPERATOR CONTEXT
Show current alarms, trends, historical tables, reports, maps, login info, and permission-aware controls.

CHARTS AND GRAPHS
Line charts, bar charts, uPlot/chart support, history loading, grouping, refresh intervals, and report chart rendering.

TABLES
History tables, alarm tables, alarm history, report tables, filters, columns, and operator review views.

MAPS AND LOCATIONS
Locations with latitude/longitude, marker settings, actions, linked views, cards, URL links, and start location.

HMI NEED	OPTIPANEL SUPPORT	EXAMPLE
Equipment page	SVG graphics, bindings, gauges, controls, permissions, alarms, and trends.	Pump faceplate with running status, speed, start/stop, trip, runtime, and last alarms.
Area overview	Multi-view pages, navigation, maps, grouped objects, charts, and dashboard values.	Production line overview, building floor, utility room, water treatment process, or remote station map.
Operational dashboard	Tables, charts, reports, alarm summary, historical values, and formatted KPIs.	Shift dashboard showing throughput, alarms, energy, downtime, and process stability.

Abnormal conditions become visible, actionable, acknowledged, and reviewable.

OptiPanel alarm services continuously evaluate tag values, expose active alarms, maintain alarm history, apply acknowledgement modes, and trigger configured actions.

ALARM PRIORITIES
High-high, high, low, information, and action alarm types support different levels of operating response.

ACKNOWLEDGEMENT
Floating, active acknowledgement, and passive acknowledgement modes with ACK user and timestamps.

ACTIONS
Set value, run script, show popup, set view, or show toast messages based on alarm configuration.

HISTORY
Alarm history includes on time, off time, ACK time, user, group, status, priority, and filtered review.

NOTIFICATION PATHS

- Email or URL/Web API notification modes.
- Subscriptions by alarm priority and timing rule.
- Delay and interval settings to avoid noisy repeated messages.
- SMTP settings and report delivery use the same operational communication model.

EVENT HISTORY

- Runtime event services record operating actions such as login, tag writes, alarms, and messages.
- Event filters and date ranges support troubleshooting and review.
- Alarm actions and scripts can guide operators to relevant pages.
- Permission filtering keeps alarm visibility aligned with operating responsibility.

CONDITION	SCADA BEHAVIOR	OPERATOR BENEFIT
Pump fault or breaker trip	Alarm appears, page can change, notification can send, ACK is recorded, history is retained.	Response is visible and documented.
Tank high-high level	Priority alarm evaluates tag limit and can trigger a script or command action if configured.	The system can guide or automate a protective response.
Communication loss	Device status, reconnect logic, logs, and alarm display can show the failed data path.	Operators distinguish process alarms from data acquisition issues.

Operational data is stored, trended, queried, and packaged into reports.

The DAQ layer records current and historical tag values. Report jobs can combine text, historical data, alarm history, and generated charts into scheduled PDF outputs.

SQLite

Local historian option with per-device storage and retention handling.

InfluxDB

Time-series storage support for InfluxDB 2.x style deployments.

Influx 1.8

Compatibility path for InfluxDB 1.8 deployments.

TDengine

Industrial time-series backend option for high-volume data.

QuestDB

Time-series backend option with fallback behavior if dependencies are unavailable.

CURRENT VALUES

Last known values are stored for device-level current readings and restoration scenarios.

HISTORICAL QUERIES

Queries support multiple tags, time ranges, chart series, table rows, and chunked Socket.IO transfer.

PDF REPORTS

Reports can include text blocks, DAQ tables, alarm history tables, generated charts, headers, footers, and email delivery.

REPORTING NEED	OPTIPANEL CAPABILITY	TYPICAL OUTPUT
Daily operating record	Scheduled report with DAQ tables, chart snapshots, and selected alarm history.	Daily plant summary for operations and management.
Incident review	Historical tag range, alarm history range, events, ACK user, and trend comparison.	Timeline of what happened before and after an abnormal condition.
Performance analysis	Trend charts, grouped graph data, history loading, tables, and report-ready data.	Energy, throughput, uptime, process stability, or equipment utilization view.

Scripts, schedules, and workflows turn SCADA data into controlled actions.

OptiPanel includes browser-side and server-side scripts, scheduling, alarm-triggered actions, system functions, and Node-RED integration so project logic can live beside the HMI and runtime model.

SCRIPT FUNCTIONS

getTag

setTag

getTagId

setView

openCard

enableDevice

getDevice

getHistoricalTags

sendMessage

getAlarms

ackAlarm

SCHEDULER

Time mode, event mode, recurring schedules, days, months, month days, all-day behavior, duration, active/disabled state, and master control enforcement.

NODE-RED

Optional flows and dashboard paths with helper nodes for tags, DAQ, alarms, views, scripts, devices, events, messages, and alarm acknowledgement.

WORKFLOW	IMPLEMENTATION PATH	USE CASE
Simulation and staging	Internal tags plus scripts and scheduler events update values, valve states, pump states, alarms, and trends.	Validate process views and operator workflows before commissioning.
Scheduled operation	Scheduler writes configured tags by time, duration, recurrence, and active period.	Building mode changes, pump rotation, test cycles, lighting, HVAC scheduling.
Alarm response	Alarm action runs a script, writes a tag, navigates to a view, or triggers notification.	Move an operator directly to the affected equipment and record the response.
External orchestration	Node-RED nodes, REST API, API keys, HTTP requests, MQTT, and ODBC connections.	Bridge SCADA with maintenance, messaging, databases, analytics, and business systems.

Access control is part of the operating model, not an afterthought.

OptiPanel includes users, groups, roles, token authentication, API keys, guest access, permission-filtered project delivery, and authorization settings for controls, alarms, scripts, gauges, and resources.

AUTHENTICATION
Sign in, JWT access tokens, refresh-token flow, configurable expiration, secure mode, and guest token behavior.

AUTHORIZATION
Users, groups, roles, admin permissions, show/enabled permission checks, and role-aware project filtering.

API ACCESS
API keys with name, value, description, creation date, expiration, enabled state, copy, generate, and permission mapping.

OPERATIONAL CONTROL
Permissions can apply to gauges, scripts, alarms, user access, frontend script functions, views, and resources.

COMMON ROLES

Viewer

Operator

Engineer

Supervisor

Manager

Administrator

- Viewer sees process state without write authority.
- Operator controls equipment and acknowledges allowed alarms.
- Engineer edits views, tags, scripts, alarms, and project settings.
- Administrator manages users, roles, API keys, and platform settings.

GOVERNANCE PRACTICES

- Separate visibility from control with show/enabled permission logic.
- Use role-based engineering pages and operator pages.
- Protect API access with generated keys and expiration strategy.
- Keep Node-RED access in secure mode when exposed to operators or engineers.

RISK	PLATFORM MECHANISM	RESULT
Unauthorized write command	Enabled permissions on HMI controls, scripts, alarms, and API paths.	Only approved roles can operate sensitive points.
Uncontrolled integration	API keys, secure authentication, and scoped project permissions.	External systems can connect without sharing human login credentials.
Unclear accountability	Alarm ACK user, events history, logs, and reportable history.	Critical actions can be reviewed after operation.

OptiPanel is extensible through plugins, REST APIs, resources, scripts, and flow automation.

A modern SCADA platform must fit into an existing environment. OptiPanel exposes a broad API surface and supports server plugins, Node-RED integration, resources, widgets, project data, command endpoints, and external notifications.

REST API AREAS
Auth, settings, project, project data, demo project, device properties, upload, users, roles, alarms, plugins, logs, resources, DAQ, scheduler, scripts, commands, reports, and API keys.

PLUGIN GROUPS
Connection device, connection database, chart image for report, and service plugins with status and install/remove handling.

RESOURCES AND WIDGETS
Resources, templates, widgets, custom SVG assets, generated resources, icons, reusable panels, and imported operating components.

EXTENSION POINT	WHAT IT ENABLES	EXAMPLE
Command API	Read values, write values, and download command-related data through controlled endpoints.	External service sets a maintenance mode tag after a work order approval.
Node-RED nodes	Read tags, set tags, get DAQ, get alarms, get history, execute scripts, open cards, emit events, and send messages.	Flow sends a message when a remote station trips and opens a detailed HMI card.
Web API device	Poll HTTP services and normalize response fields into OptiPanel tags.	Read values from a cloud metering endpoint or vendor service.
Database connection	ODBC and Redis allow integration with operational databases, caches, and edge systems.	Display production plan data beside live line status.

Extensibility is part of the product position: OptiPanel can supervise a process directly, connect to existing data systems, expose selected functions to other tools, and automate workflows without forcing every requirement into a single HMI screen.

Deploy on a server, package as desktop, run in Docker, or operate headless.

Portability is a key part of OptiPanel. The same platform can be delivered as a web server for browser sessions, a containerized runtime, a desktop package, or a headless process depending on the site and project requirements.

LOCAL SERVER
Node.js backend with Angular frontend, default server port pattern, runtime directories, logs, reports, and project storage.

DOCKER
Dockerfile based on Node 18 with mounted app data, database, and logs directories for repeatable deployment.

DESKTOP
Electron package copies the server, dependencies, and built frontend into a Windows, Linux, or macOS application.

HEADLESS
Headless entry point stores user data under a home-directory data folder and launches the runtime server.

SMALL SYSTEM
Single machine, local server, SQLite historian, browser HMI, simple users, and local reports.

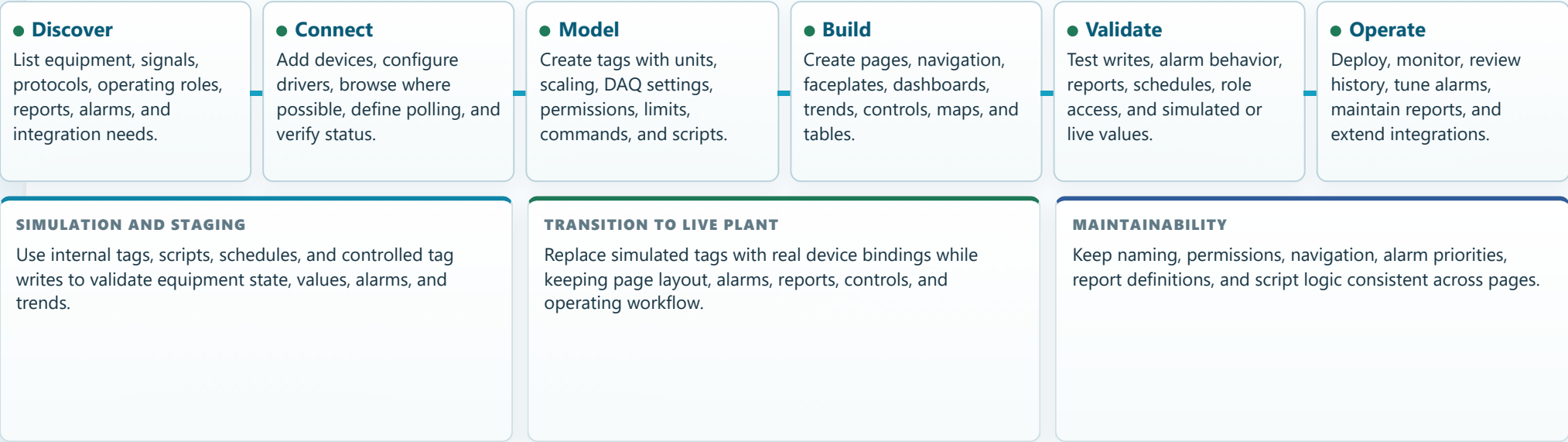
PLANT SYSTEM
Multiple devices, role-based screens, alarms, notifications, historian backend, reports, and API keys.

DISTRIBUTED SYSTEM
Remote stations, MQTT/HTTP/OptiPanel Server links, Node-RED workflows, maps, notifications, and container deployment.

DEPLOYMENT CHOICE	BEST FIT	OPERATIONAL CONSIDERATION
Server plus browser	Control rooms, engineering laptops, operator stations, tablets on secured networks.	Central runtime with multiple browser sessions and role-specific access.
Docker	Repeatable deployment, infrastructure-managed runtime, backup/restore discipline.	Mount data, database, and logs volumes intentionally.
Electron desktop	OEM delivery, local operator workstation, packaged demo, offline engineering showcase.	Frontend and server are bundled into a desktop-oriented package.

From process discovery to maintainable SCADA application.

OptiPanel projects should be engineered around the process: equipment, signals, operating modes, control authority, alarms, history, reports, and lifecycle support.



ENGINEERING ASSET	EXPECTED CONTENT	OPTIPANEL AREA
Point list	Tag ID, device address, type, unit, range, DAQ, alarm limits, write permission, and HMI display rule.	Connections, tag properties, DAQ, alarms, permissions.
HMI page map	Overview, area pages, equipment pages, alarms, trends, reports, maps, and settings access.	Views editor, layout, navigation, cards, maps, tables.
Operating logic	Schedules, scripts, alarm actions, mode commands, startup checks, and simulated demo behavior.	Scripts, scheduler, alarm actions, command functions, Node-RED.
Handover package	Deployment mode, users, roles, API keys, backup paths, reports, historian storage, and maintenance notes.	Settings, users, roles, API keys, reports, DAQ storage, deployment.

Rich feature coverage across the SCADA lifecycle.

OptiPanel combines the elements needed to build, operate, extend, secure, and maintain process visualization and supervisory control applications across industries.

CATEGORY	CAPABILITIES	REPRESENTATIVE MODULES
Connectivity	OPC UA, S7, Modbus TCP/RTU, BACnet, MQTT, HTTP, ODBC, ADS, PCCC, GPIO, webcam, MELSEC, Redis, OptiPanel Server.	Device manager, plugins, protocol drivers, device property APIs.
HMI engineering	Views, SVG editor, charts, graphs, tables, maps, gauges, controls, custom styles, layout, navigation, resources, widgets.	Frontend editor, visual libraries, resources, layout settings.
Operations	Alarms, ACK, alarm history, event history, notification, scheduler, scripts, report jobs, live command functions.	Alarm manager, events history, noticator, scheduler, scripts, jobs.
Data and reporting	Current values, DAQ historian, SQLite, InfluxDB, InfluxDB 1.8, TDengine, QuestDB, charts, tables, scheduled PDFs.	DAQ storage, current storage, report generator, chart/report plugins.
Security	Users, roles, groups, JWT, refresh tokens, guest access, API keys, authorization on controls, scripts, alarms, and resources.	Auth API, users runtime, API keys runtime, permission checks.
Integration	REST API, Node-RED, Web API devices, MQTT, ODBC, Redis, notifications, scripts, command endpoints, resources APIs.	OpenAPI routes, Node-RED integration, plugin registry, API keys.
Portability	Node.js server, Angular web interface, Docker, Electron package, headless runtime, local folders for app data, DB, logs, and reports.	Server, client distribution, Dockerfile, Electron package, headless entry.

PRODUCT STATEMENT

OptiPanel is a portable and extensible SCADA, HMI, dashboard, historian, automation, and integration platform for sensor-actuator processes.

ENGINEERING STATEMENT

Model tags once, then reuse them in HMI pages, alarms, historian, reports, scripts, schedules, APIs, and Node-RED flows.

OPERATING STATEMENT

Give operators a central place to monitor equipment, control permitted devices, acknowledge alarms, review history, and understand system state.

The water treatment HMI is one example application. The platform is broader: any industrial, utility, facility, or machine process with measurable state and controllable equipment can be engineered as an OptiPanel SCADA application.